

ULMS Debugging Evidence

Task 5: Debugging Evidence - Summary

The following defects were identified in the supplied faulty modules and corrected in the included source files.

Debugging Evidence Table

Defect	Original Faulty Code	Issue	Corrected Code	Explanation
BUG-001 Login	<code>if (username == "admin" password == "1234")</code>	OR allowed partial credentials to login	<code>if (username == "admin" && password == "1234")</code>	Both username and password must be correct for authentication.
BUG-002 Average	<code>record.Average = record.Subject1 + record.Subject2 + record.Subject3 / 3;</code>	Division happened before addition	<code>record.Average = (record.Subject1 + record.Subject2 + record.Subject3) / 3.0;</code>	Parentheses ensure all three subjects are summed before division.
BUG-003 Marks input	<code>Convert.ToDouble(txtSubject1.Text)</code>	Blank or text input crashes application	<code>double.TryParse(..., out double subject1)</code>	TryParse allows validation error instead of unhandled exception.
BUG-004 Enrolment	No duplicate check before output	Same course can be enrolled repeatedly	Check existing enrolments with <code>Any(...)</code> before saving	Business rule prevents duplicate course registrations.
BUG-005 Reports	<code>Thread.Sleep(4000);</code>	Creates unacceptable delay	Removed <code>Thread.Sleep</code> and generated report immediately	Improves performance and meets report response requirement.
BUG-006 Age	<code>Age = int.Parse(txtAge.Text)</code>	Non-numeric age crashes application	<code>int.TryParse(txtAge.Text, out int age)</code>	Displays clear error instead of crashing.